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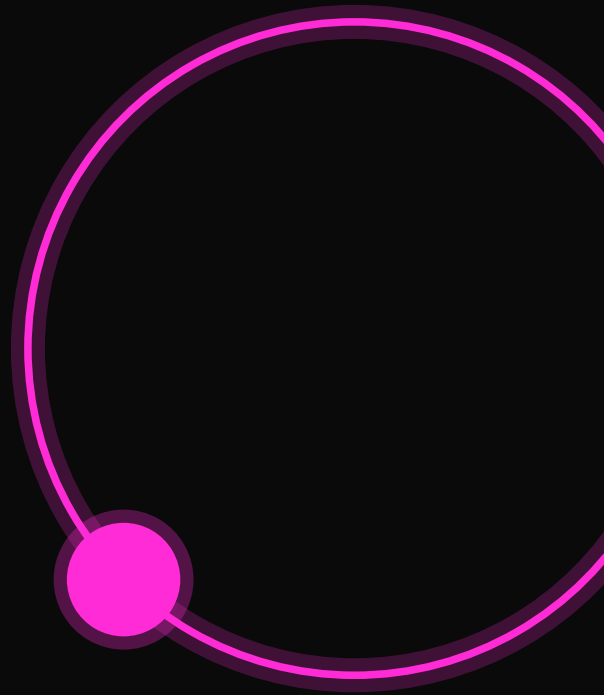
Dialogue Systems 2

Department of Philosophy, Linguistics and
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E.L.I.S.A.

Entertainment **L**ookup with
Information-**S**tate **A**rchitecture



A spoken movie assistant, powered by SISU

`your popcorn-powered movie wingman`

Two phases

1 Phase 1: form filling

- Who is watching & the youngest age
- Time limit & genres to skip
- Country & streaming services

These are hard filters. They decide tonight's pool of movies.

2 Phase 2: chatting

- Converse about mood and preference
- Elisa suggests from the pool
- Repeat until you pick a movie

Spoken, with Azure. English only.

Hard facts are clicked, not said, so they cannot be misheard.

TMDB: the movie data

A free movie database with genres, runtimes, ratings, keywords, cast and streaming availability.

~23,000

movies in the local catalogue, each with a vector

500 to 1,300

movies in a typical Phase 1 pool

0

TMDB calls during the chat

From moods to wishes

Version 1

- Mood word to genre lookup table
- Genre was a hard TMDB filter
- About 60 results re-ranked, descriptions embedded live
- A wrong genre guess hid the right movie

Version 2 (now)

- Phase 1 filters only, then rank the whole pool by meaning
- Vectors precomputed for ~23,000 movies
- Only the wish is embedded at chat time
- The chat becomes a wish plus an avoid list

Building blocks

speechstate

Azure speech in and out. Strict turn-taking.

xstate

Top machine: form, pool, chat.

SISU

Information state plus update rules.

NLU (LLM)

Utterance to moves: wish, avoid, person, seen, reject, accept.

NLG (LLM)

Writes Elisa's reply. Code picks the mode: explore, recommend, discuss or wrap-up.

Embeddings

Wish to vector, then rank the pool.

TMDB

The pool and the facts. *No invented plots.*

Code guards

One example: Grounding drops moves the user never said.

How ranking works

score = **0.8 meaning** + **0.1 popularity** − **avoid** − **reject**

Meaning

How close the wish (or a liked movie) is to each movie's vector, as a percentile in the pool.

Avoid

Only the 5% most similar to the avoided thing. "No princesses" embeds like "princesses", so it is kept out of the wish.

Reject

The 10% most similar to a turned-down movie sink.

Three models, three jobs

NLU

University llama3.1

Then Groq, then local qwen2.5:7b

NLG

Groq gpt-oss-120b

Then university llama3.1, then local qwen2.5:7b

Embeddings

Local qwen3-embedding:4b

2,560 dimensions

Strongest raw similarity of those tried

- The 4B embedder fills the GPU, so the LLM moved to the university server
- Groq has daily and per-minute limits, so it is NLU's fallback and NLG's first choice
- Benches: nlu-bench 31 of 32 cases on university llama3.1, rank-bench recall@10 52%

1st Dialogue: family evening

Illustrative dialogue. Phase 1 setup: family, youngest 8, 100 minutes.

What is said	What the system does
Elisa: Hi Sam! What are you all in the mood for tonight?	Fixed opening, no LLM
You: Something funny for the kids, but no princesses.	wish: a funny family movie. avoid: princesses
Elisa: How about Chicken Run? A clever escape comedy, 84 minutes.	Pool re-ranked, top movie suggested
You: Not that one. Something with more adventure.	reject: Chicken Run. wish rewritten: a funny family adventure
Elisa: Then try Wallace & Gromit: The Curse of the Were-Rabbit.	Look-alikes of Chicken Run sink
You: Yes, let's watch that!	A clear yes becomes accept, no LLM call

2nd Dialogue: date night

Illustrative dialogue. Phase 1 setup: couple, no time limit.

What is said	What the system does
Elisa: Hi Alex! What are you two in the mood for?	Fixed opening, no LLM
You: A romantic comedy, something light.	wish: a light romantic comedy
Elisa: Crazy, Stupid, Love is a warm, funny one.	Top-ranked movie suggested
You: Seen it. Anything with Emma Stone?	seen: removed from the pool. person: Emma Stone, ranked first
Elisa: Easy A: sharp and funny, with Emma Stone in the lead.	Person boost applied
You: That sounds perfect.	accept, then Elisa wraps up

Problems, limits, future work

Problems

- Elisa sometimes names a movie without recording it
- Small models echo Elisa's words as the user's
- The NLU can still miss a wish

Limits

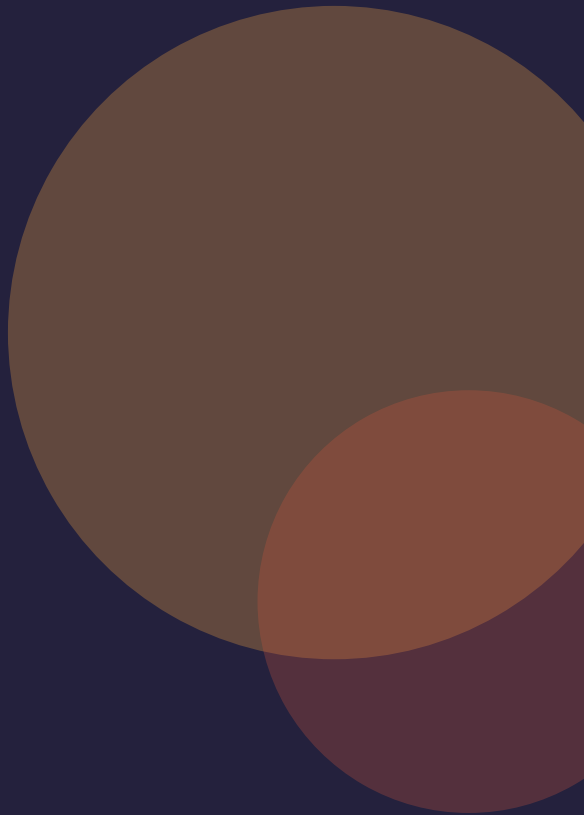
- English only
- Year or runtime changes go back to the form
- Groq limits slow live chats

Future work

- **User testing, first of all**
- Barge-in while Elisa speaks
- Stronger actor and director handling
- Better tuned NLU

Demo

The form, a wish, a reject, a seen movie, an accept



Questions?

SISU · xstate · speechstate · embeddings

